12.6.4 Wrap-Up: Cool Down

1. Write **one question** you still have about solving mathematical and real-world problems involving the lengths of secants, tangents, segments, or chords of a circle.

Unit 12, Lesson 7: Deriving the Equation of a Circle

Benchmark

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.

Learning Targets

- ☐ I can use the Pythagorean Theorem to derive the equation of a circle.
- ☐ I can determine key features of a circle given the graph or equation.
- ☐ I can write the equation of a circle using key features.

12.7.1 Warm-Up: Roundabouts

Roundabouts promote a continuous flow of traffic because drivers do not have to wait for a green light at a roundabout to get through the intersection. A diagram of a roundabout is shown.

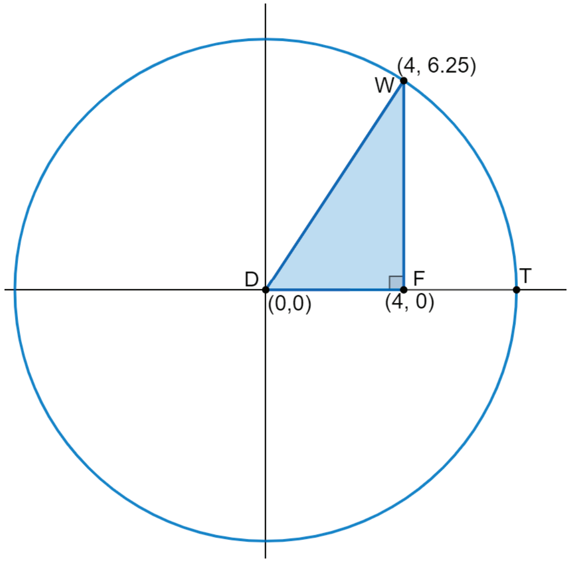
1. What do you wonder about the diagram?

2. How many different paths could the red car take?

3. List at least two things you have studied in Geometry so far that may be helpful to know when constructing a roundabout.

12.7.2 Exploration: Deriving the Equation of a Circle

1. The center of circle D is located at $(0,0)$. Point W lies on circle D at the point $(4, 6.25)$. Right triangle DFW is drawn inside the circle, such that $\overline{DW}$ is a circle's radius and point F, $(4,0)$, lies on radius $\overline{DT}$.



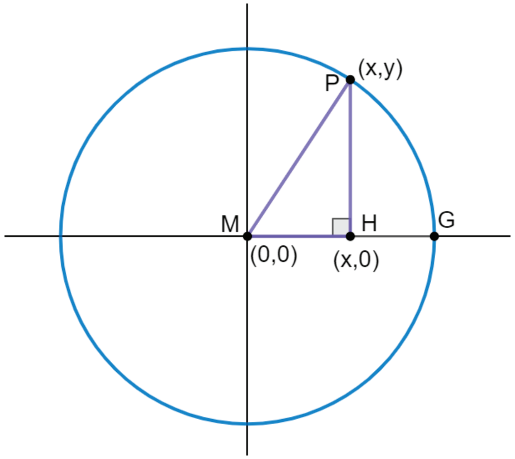
a. Find the length of each side.

Line Segment	Distance Between Endpoints
$\overline{DF}$	
$\overline{FW}$	

b. Use the values you found for the lengths of $\overline{DF}$ and $\overline{FW}$ to complete the equation that can be used to determine the length of $\overline{DW}$ by applying the Pythagorean Theorem.

$(DW)^2 = (\qquad)^2 + (\qquad)^2$

2. The center of circle M is located at $(0,0)$. Point P lies on circle M at the point (x, y). Right triangle MPH is drawn inside the circle, such that $\overline{MP}$ is a circle's radius and point H, $(x, 0)$, lies on radius $\overline{MG}$.



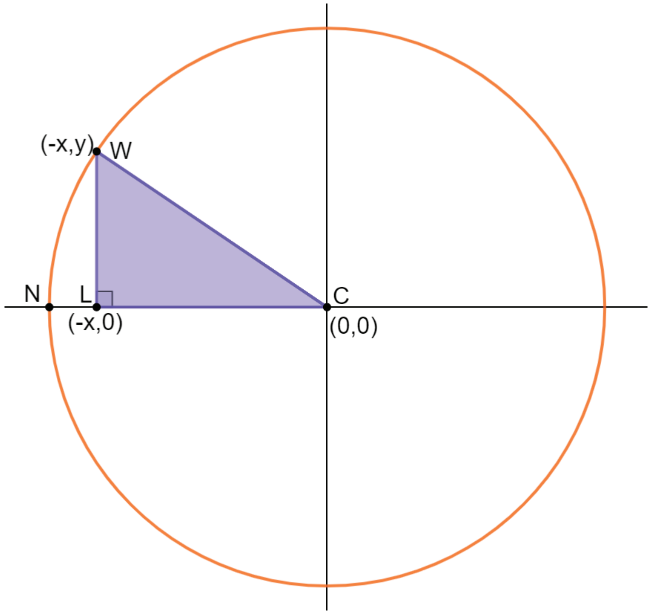
a. Write an expression that represents the length of each side.

Line Segment	Distance Between Endpoints
$\overline{MH}$	
$\overline{HP}$	

b. Use the expressions you found for the lengths of $\overline{MH}$ and $\overline{HP}$ to complete the equation that can be used to determine the length of $\overline{MP}$ by applying the Pythagorean Theorem.

$(MP)^2 = (\qquad)^2 + (\qquad)^2$

3. The center of circle C is located at $(0,0)$. Point W lies on circle C at the point $(-x, y)$. Right triangle WLC is drawn inside the circle, such that $\overline{WC}$ is a circle's radius and point L, $(-x, 0)$, lies on radius $\overline{NC}$.



a. Write an expression that represents the length of each side.

Line Segment	Distance Between Endpoints
$\overline{LC}$	
$\overline{LW}$	

b. Use the expressions you found for the lengths of $\overline{LC}$ and $\overline{LW}$ to complete the equation that can be used to determine the length of $\overline{WC}$ by applying the Pythagorean Theorem.

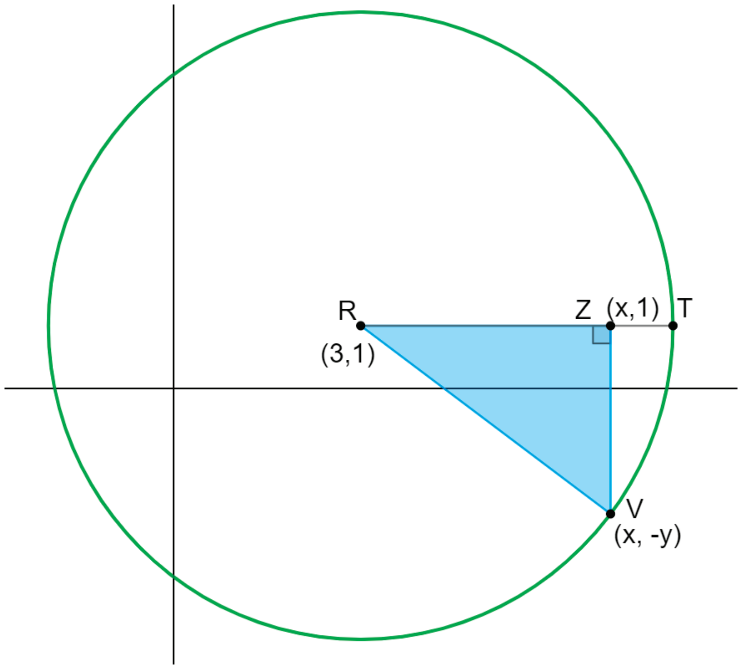
$(WC)^2 = (\qquad)^2 + (\qquad)^2$

c. Complete the table with the equation you found that can be used to determine the length of the radius of each circle.

Line Segment	Distance
$\overline{MP}$ radius of circle M	$(MP)^2 =$ or $(r)^2 =$
$\overline{WC}$ radius of circle C	$(WC)^2 =$ or $(r)^2 =$

In circle M and circle C, the location of the point on the circle was in a different location, but the equations that can be used to determine the length of the radius of the circles are the same.

4. The center of circle R is located at $(3,1)$. Point V lies on circle R at the point $(x,-y)$. Right triangle RZV is drawn inside the circle, such that $\overline{RV}$ is a circle's radius and point Z, $(x,1)$, lies on radius $\overline{RT}$.



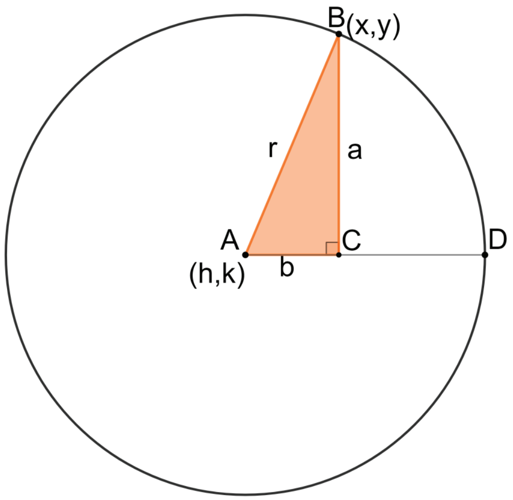
a. Write an expression that represents the length of each side.

Line Segment	Distance Between Endpoints
$\overline{RZ}$	
$\overline{ZV}$	

b. Complete the equation that can be used to determine the length of the radius of circle R.

$(r)^2 = (\qquad)^2 + (\qquad)^2$

5. The center of circle A is located at (h, k). Point B lies on circle C at the point (x, y). Right triangle ABC is drawn inside the circle such that $\overline{AB}$ is a circle's radius and point C, lies on radius $\overline{AD}$. $AC = b$, $BC = a$, and $AB = r$.



a. Use (x, y) and (h, k) to determine the ordered pair for point C.

b. Fill out the table by using the ordered pairs to complete each equation.

Line Segment	Distance Between Endpoints
$\overline{AC}$	$b =$
$\overline{BC}$	$a =$

The Pythagorean Theorem can be used to write the equation, $a^2 + b^2 = r^2$.

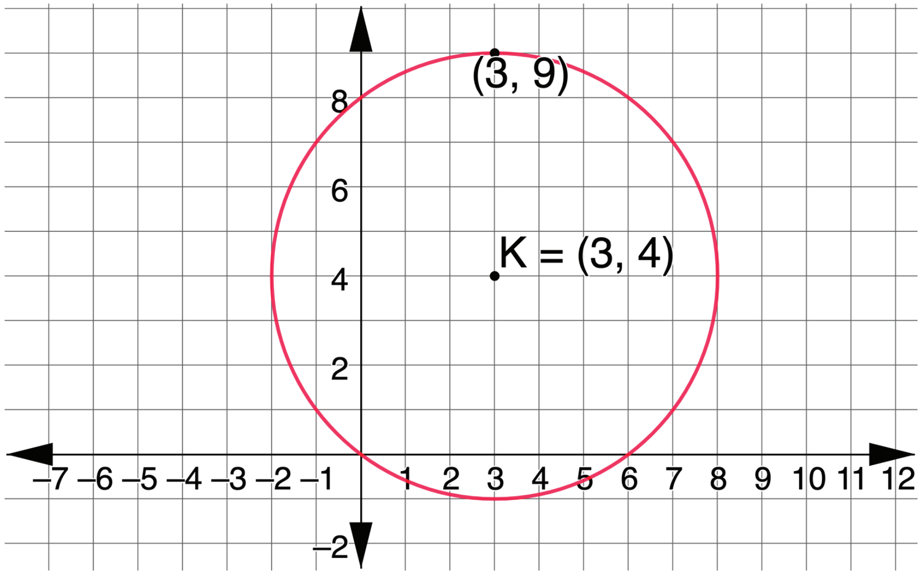
c. Use the equations in the table to substitute for a and b in the equation $a^2 + b^2 = r^2$.

The **center-radius form** of an equation of a circle with center, (h, k), and radius, r, is $(x - h)^2 + (y - k)^2 = r^2$.

12.7.3 Guided Instruction: Key Features of a Circle

Circles have two key features: the center and the radius. The key features can be used to write the equation, $(x-h)^2+(y-k)^2=r^2$, of the circle.

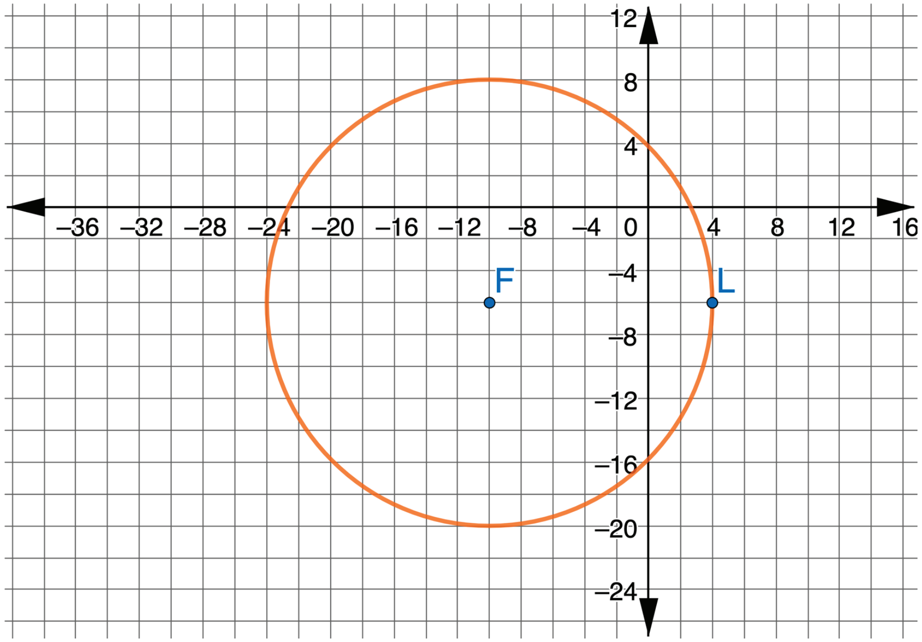
1. Circle K is shown with center $(3,4)$. The point $(3,9)$ is on circle K.



   a. Determine the length of circle K's radius.

   b. Write the equation that represents circle K.

2. Circle F is shown, where point L is on the circle.



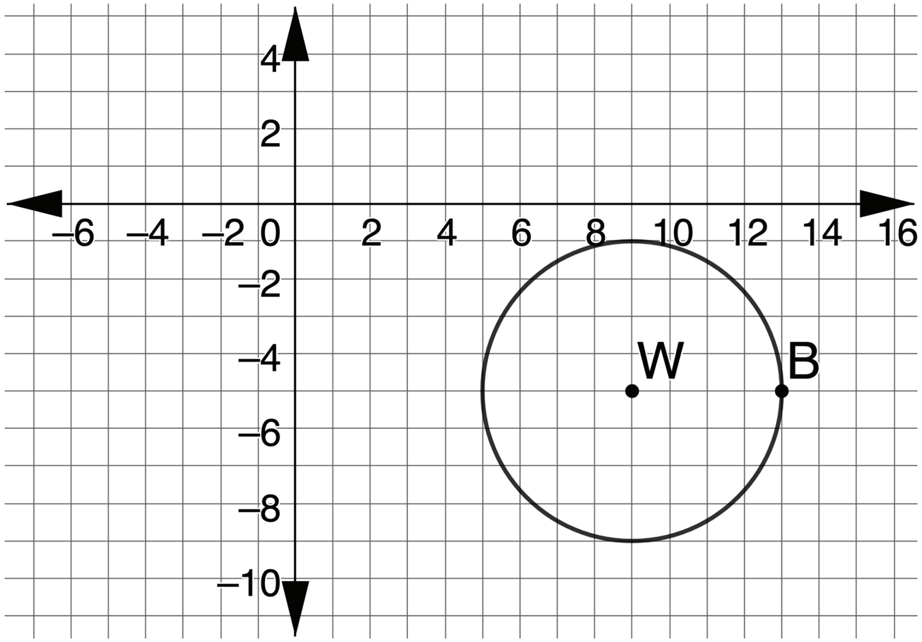
   a. Determine the key features of circle F.

Key Features	
Radius	
Center	

   b. Write the equation that represents circle F.

12.7.4 Wrap-Up: Ticket Out the Door

1. Circle W is shown, where point B is on the circle.



a. Determine the key features of circle W.

Key Features	
Radius	
Center	

b. Write the equation that represents circle W.

Unit 12, Lesson 8: Standard Form of a Circle

Benchmarks

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.

MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.

Learning Targets

- ☐ I can rewrite the equation of a circle from standard form to center-radius form.
- ☐ I can determine key features of a circle given the equation in standard form.
- ☐ I can write the domain and range using inequality notation, interval notation, or set-builder notation.